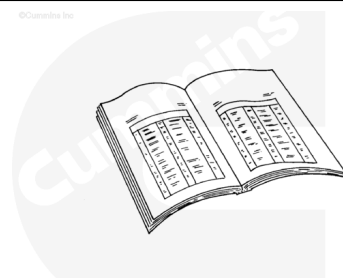


Trainings ▾

Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



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⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.

⚠ WARNING ⚠

Use caution when draining oil or replacing filters that the oil is not spilled or drained into the bilge area. The oil and oil filters must be disposed of in accordance with local environmental regulations.

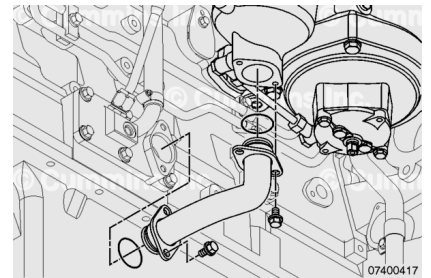
- Disconnect the battery cables. Refer to Procedure 013-009 in Section 13.
(/qs3/pubsys2/xml/en/procedures/99/99-013-009.html)
- Drain the lubricating oil from the engine. Refer to Procedure 007-037 in Section 7.
(/qs3/pubsys2/xml/en/procedures/20/20-007-037.html)

Note : It is suggested as good service practice to perform a pressure drop test on the Eliminator™ as shown further on in the document under Pressure Test. This shows how the Eliminator™ has been performing prior to removal and is a good reference check with replacing or installing, potentially after repair or inspection.

Remove

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

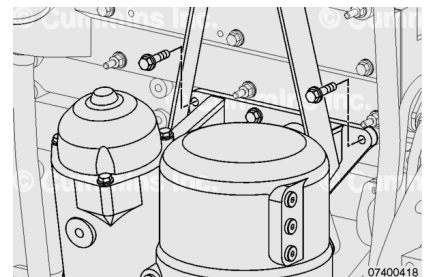


Wait for 10 minutes after shutting down prior to removing the Eliminator™ (1) to make sure the centrifuge has come to a complete stop.

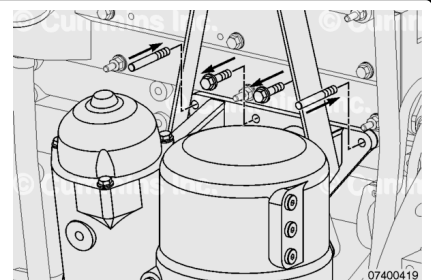
Remove the oil drain tube by removing the two capscrews that secure the tube to the filter housing and the two capscrews that secure the tube to the block.

Place a lifting strap around the filter housing to secure it before removal.

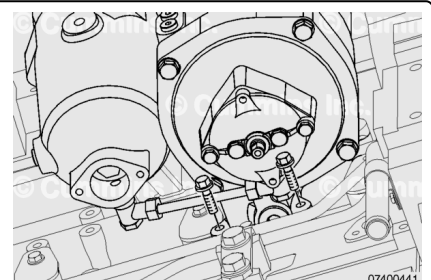
Remove the two outermost mounting capscrews at the top of the filter housing.



Install the two guide pins and remove the remaining capscrews at the top of the filter head.

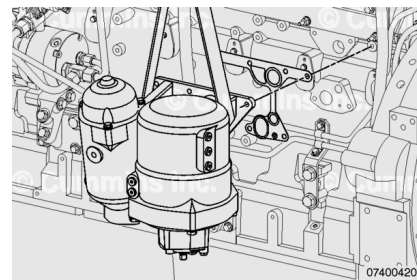


Remove the two support capscrews at the bottom of the unit.



Use the lifting strap to remove the filter assembly from the engine.

Discard the gasket.



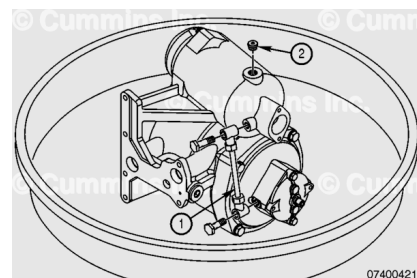
Disassemble

Place the entire unit in a shallow container to drain the oil and to keep the unit clean.

Oil may flow out of the openings at the rear of the filter head, as well as the drain opening at the bottom.

Remove the hydraulic motor return line (1) from the rear of the filter by removing the banjo fittings.

Remove the drain plug (2) on the side of the filter head to drain any oil that may remain around the filter elements.

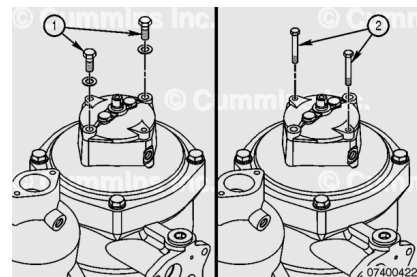


Remove the two capscrews (1) and washers from the hydraulic motor.

Insert two M10 capscrews (2) on the opposite threads of the hydraulic motor.

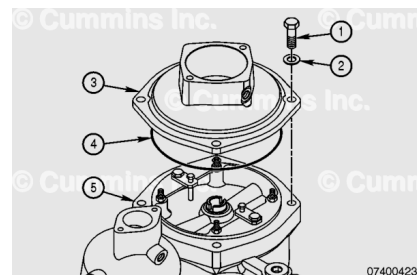
Alternately tighten each capscrew one half of a turn to assist in pulling the motor from the housing.

Lift the hydraulic motor out of the housing.



Remove the four capscrews (1) and washers (2) from the filter head (3).

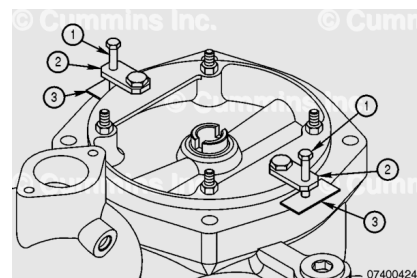
Remove the filter head (3) from the housing (5) and discard the o-ring (4).



Loosen the extraction system by removing the two capscrews (1).

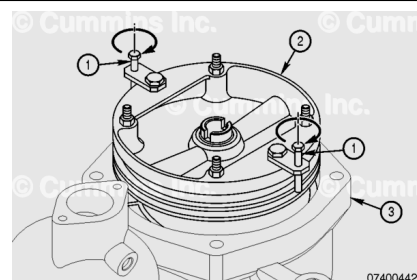
Rotate the plates (2) one half of a turn so they face outboard of the centrifuge.

Place a small plate (3) or piece of sheet metal on the housing under the two capscrews (1) to protect the aluminum housing.

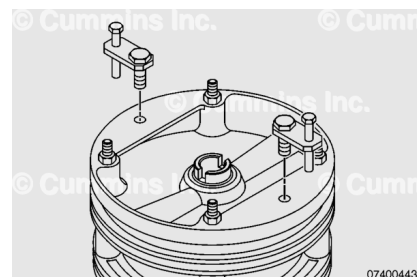


Tighten the two capscrews (1) to lift the filtering unit (2) slightly out of the filter body (3).

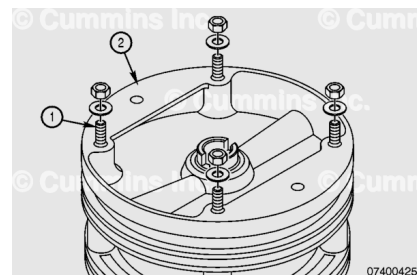
Hold the filtering unit (2) by the two handles on the top and lift it out of the filter body (3).



Remove the extraction system from the filtering unit.

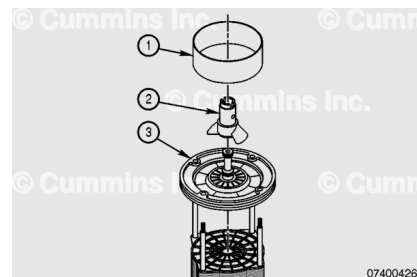


Remove the four nuts and washers from the rods (1). Remove the cover (2).



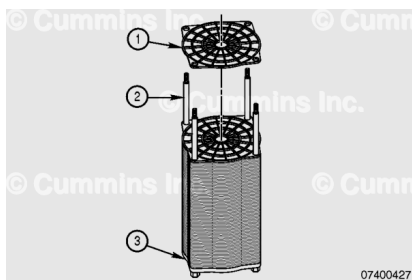
Remove the mesh strainer (1).

Remove the distributor (2) and the distribution cover (3).



Remove the filtering elements (1) one at a time from the rods (2).

Remove the rods (2) from the base (3).

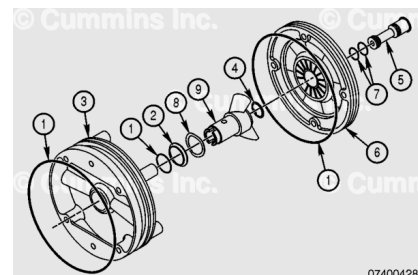


Remove the o-rings (1) and bearing (2) from the cover (3).

Remove the o-ring (1), circlip (4), and the shaft (5) from the distribution cover (6).

Remove the two o-rings (7) from the shaft (5). Remove the spring washer (8) from the distributor (9).

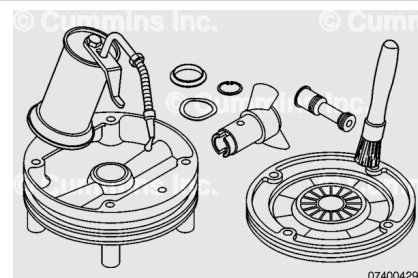
Discard all o-rings.



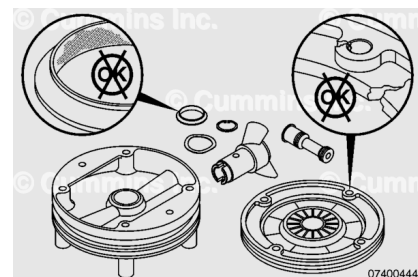
Clean and Inspect for Reuse

Note : Follow these steps for all pieces except for the filter elements.

Use a soft brush and carefully clean all components with clean lubricating engine oil.

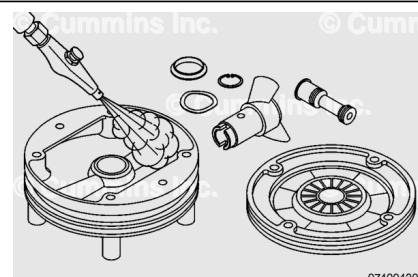


Inspect the components. Replace if they are worn or damaged.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

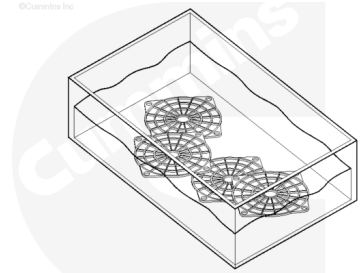


Use compressed air to remove any debris from the components.

⚠ WARNING ⚠

When using solvents, acids or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

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07400388

⚠ CAUTION ⚠

Do not use acid-based cleaning solvents. This can damage the filter elements.

Note : Follow these steps for the filter elements only.

Keep one full-flow filter element as a guide for assembly of the filter halves after cleaning.

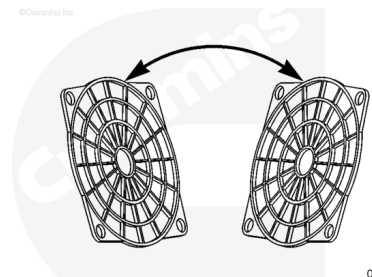
Put the filter elements into a tray, or equivalent, filled with cleaning solvent.

Soak the filter elements in the tray for at least 15 minutes.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

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⚠ CAUTION ⚠

Rinse thoroughly with water. Engine damage can result if solvent is left on the filter elements.

Remove an element from the cleaning tray and separate it into halves.

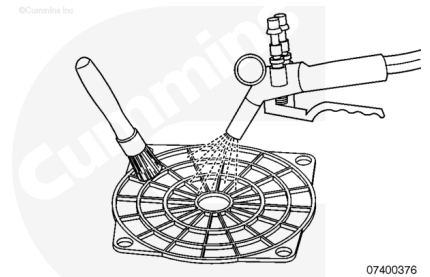
⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

⚠ CAUTION ⚠

Rinse thoroughly with water. Engine damage can result if solvent is left on the filter elements.

Brush the element halves with a soft brush. Use a hot water spray to assist the process.

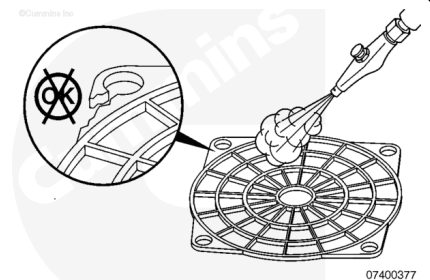


Dry the element halves with compressed air.

Carefully inspect the filter elements for damage or dirt.

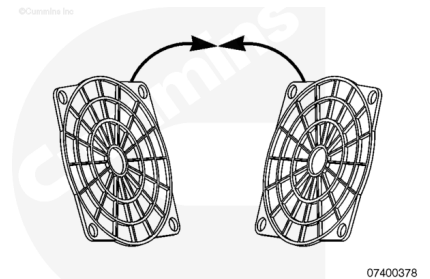
Always check the filter screen for cracks. Replace any damaged elements.

If the elements are still dirty, repeat the cleaning procedure. Increase the time the elements are in the cleaning tray.



Assemble the element halves, place them on a clean surface, and repeat the steps for the elements remaining in the solvent.

When all elements have been cleaned and assembled in the same manner as the one saved for reference, clean the reference element.



Assemble

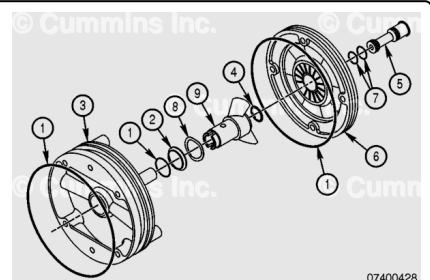
Replace the o-rings (1) with new o-rings.

Lubricate the o-rings with clean lubricating engine oil and fit them on the cover (3).

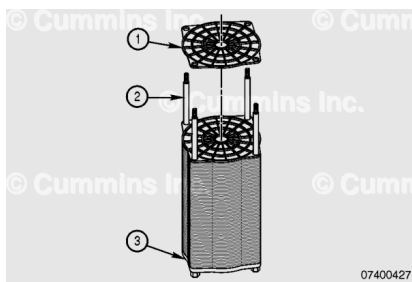
Install the bearing (2) into the cover (3). Install the circlip (4).

Lubricate the shaft (5) with clean lubricating engine oil and place it in the distribution cover (6).

Place two new, lubricated o-rings on the shaft (7). Install a new spring washer (8) on the distributor (9).

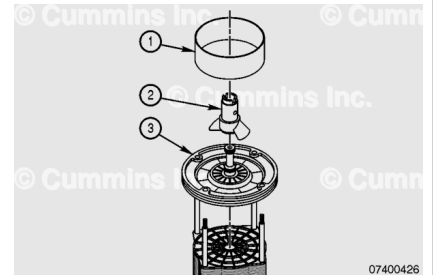


Place the rods (2) back onto the base (3) and place each of the clean filtering elements (1) over the rods, one at a time.



Lubricate the distributor (2) with clean lubricating engine oil and place it onto the shaft of the distribution cover (3).

Place the strainer (1) on top of the distribution cover (3).



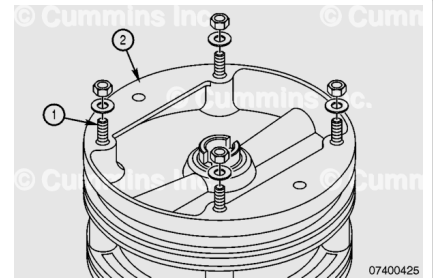
Place the cover (2) on top of the strainer.

Place the assembly on top of the rods.

Place the washers and nuts (1) on the rods.

Tighten the nuts.

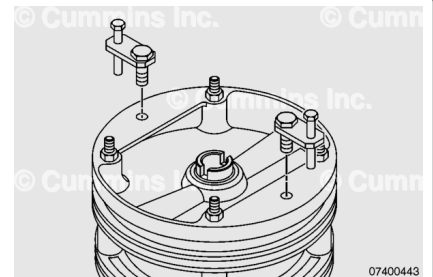
Torque Value: 10 n·m [89 in-lb]



Install the extraction system onto the cover.

Rotate the plates of the extractor half of a turn.

Hand tighten the two capscrews to lock them into position.



Hold the filtering unit by the handles on the top and insert it carefully back into the filter body.

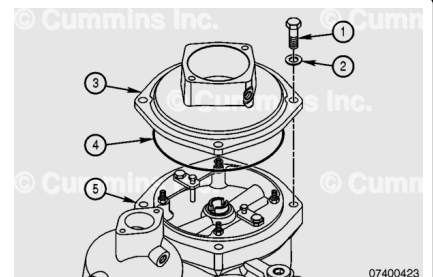
Insert the new o-ring (4).

Replace the filter head (3) on the housing (5).

Insert and tighten the four capscrews (1) with washers (2) on top of the filter head.

Tighten the capscrews.

Torque Value: 47 n·m [35 ft-lb]

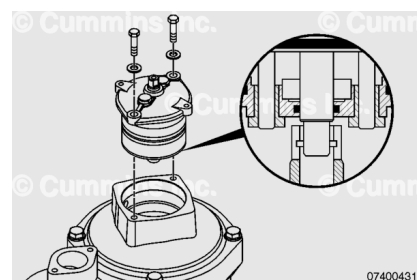


Insert the hydraulic motor back into the filter head. The driving pin must be in the correct position, corresponding to the groove in the distributor.

Insert and tighten the two capscrews and washers mounting the hydraulic motor onto the filter assembly.

Tighten the capscrews.

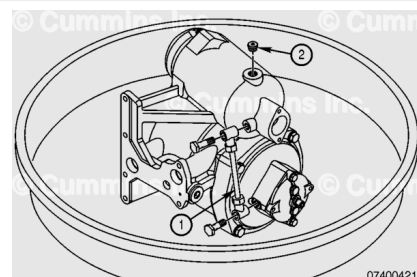
Torque Value: 27 n•m [239 in-lb]



Connect the hydraulic motor return line.

Insert and tighten the banjo fitting (1) and coupling capscrews on the hydraulic motor.

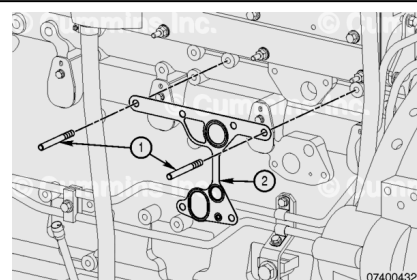
Install the drain plug (2) on the side of the filter head.



Install

⚠ WARNING ⚠

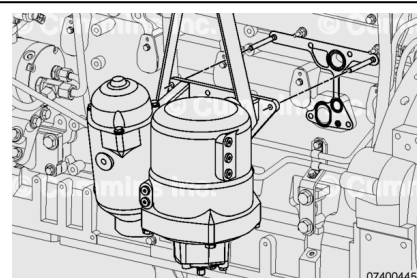
This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Install two guide pins (1) into the engine block.

Place the gasket (2) over the guide pins and align it against the block.

Place the Eliminator™ over the guide pins and push it against the engine block.



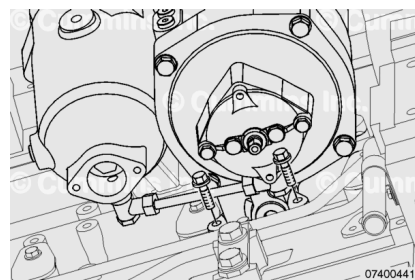
Insert the capscrews in the lower mounting holes of the flange on the lubricating oil filter head Eliminator™. Finger tighten **only**.

The gasket **must** still be in place. If **not**, loosen the capscrews and align.

If the gasket has been damaged due to misalignment, it **must** be replaced.

Tighten the two capscrews.

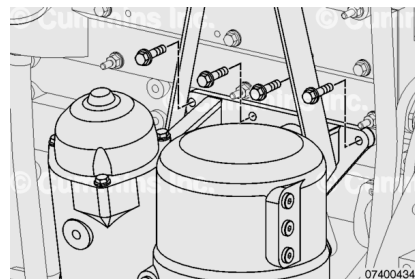
Torque Value: 27 n•m [239 in-lb]



Remove the guide pins and start the remaining four capscrews through the mounting holes in the top of the filter head. The gasket **must** be aligned.

Tighten all six capscrews.

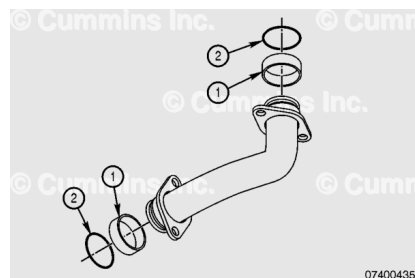
Torque Value: 41 n•m [30 ft-lb]



Assemble the drain tube with a wear sleeve (1) over each end.

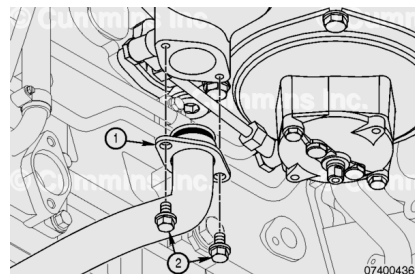
Install an o-ring (2) in the groove at each end of the drain tube.

Lubricate the o-rings with grease, Part Number 3163086, 3163087, or equivalent.



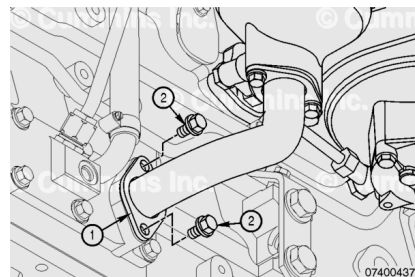
Push the top end of the drain tube into the bottom of the filter head.

Position the mounting flange (1) at the bottom of the filter head and start the two captive washer capscrews (2).



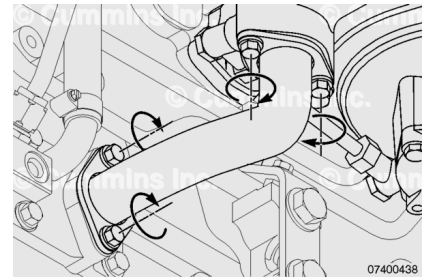
Push the bottom end of the oil drain tube into the hand hole cover and position the mounting flange (1) to align it with the mounting capscrew holes.

Finger start the two captive washer capscrews (2) through the flange of the oil drain tube and into the hand hole cover.



Tighten the four capscrews.

Torque Value: 41 n•m [30 ft-lb]



Pressure Test

Perform the following steps to measure the pressure drop across the Eliminator™ filter.

Remove one plug from each side at the top of the filter housing.

Install the pressure gauges in the ports.

Operate the engine to normal operating temperature.

Record the readings from both gauges.

The difference between the two measurements is the pressure drop across the filter.

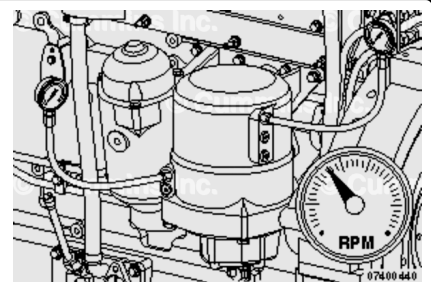
The maximum operating pressure drop **must not** exceed 39.9 kPa [5.8 psi]. If the pressure drop is greater, check the indicator on the bottom of the hydraulic motor to be sure it is turning. It **must** make one complete rotation once in 5 minutes at idle. It will **not** turn continuously, but will turn and stop. If it is **not** turning, shut the engine down. Replace the motor.

If the motor is replaced, start the pressure test again and record the pressure drop for future reference.

Remove the pressure gauges when the pressure drop value is acceptable.

Install and tighten the plugs.

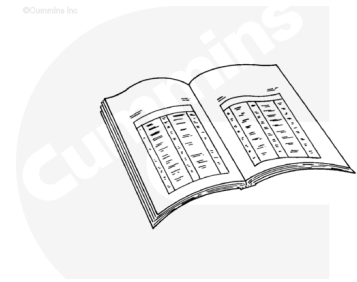
Torque Value: 10 n•m [89 in-lb]



Finishing Steps



Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



ck800wa

- Fill the engine with lubricating oil. Refer to Procedure 007-037 in Section 7.
(/qs3/pubsys2/xml/en/procedures/20/20-007-037.html)
- Connect the battery cables. Refer to Procedure 013-009 in Section 13 (/qs3/pubsys2/xml/en/procedures/99/99-013-009.html)
- Operate the engine. Check for leaks.
- While the engine is operating, check the indicator on the bottom of the hydraulic motor to be sure it is turning. It should make one complete rotation once in 5 minutes at idle. It will **not** turn continuously, but will turn and stop. If it is **not** turning, shut the engine down and replace the motor.